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**Predicting the Relationship Between the Spatial Organization of
Animal Farms and Water Quality in the Maumee River Watershed**

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temperature, NNI, phosphorus

Abstract

Animal agriculture in the United States (US) continues to industrialize and expand, transforming agricultural landscapes and raising water quality concerns, particularly in northwest Ohio. While animal feeding operations (AFOs) tend to cluster spatially, a phenomenon linked to increased nutrient loading nationwide, there is a critical need to evaluate how AFO clustering influences water quality at regional watershed scales. In this study, we analyzed the local-scale impacts of AFO spatial clustering on water quality across the Maumee River Watershed (MRW). Using animal farm locations and a 20-year water-quality dataset (2000–2020), we quantified AFO spatial clustering using a nearest-neighbor index and applied linear mixed-effects models (LMMs) to assess its influence on total phosphorus (TP), total Kjeldahl nitrogen (TKN), and water temperature (T_w). Overall, HUC10-level watersheds with clustered AFO patterns were associated with elevated nutrient concentrations and stream temperatures. Watersheds with clustered AFOs had on average, 0.86 mg/L higher TP and 0.46 mg/L higher TKN stream concentrations than watersheds with dispersed AFOs. Nutrient concentrations peaked in clustered watersheds during the summer, with mean TP and TKN concentrations of 2.05 mg/L and 1.81 mg/L, respectively, consistent with periods of increased manure application. Watersheds with clustered AFOs also exhibited T_w averaging 1.64 °C higher than dispersed watersheds and experienced warmer fall and winter stream conditions. LMMs revealed that omitting spatial configuration limits model accuracy. Including spatial pattern as a fixed effect explained 31% and 68% of variance in TP and T_w , respectively. While environmental predictors such as percent wetlands, croplands, and forest improved model performance, land cover composition alone was insufficient to explain observed water quality patterns. Regional research has focused heavily on the effects of row-crop agriculture and isolated manure application, overlooking the landscape-

scale effects of AFO spatial organization. Our findings show that incorporating AFO spatial patterns into watershed-scale assessments may improve predictions of water quality outcomes and can inform management strategies to build resilience in Great Lakes coastal systems.

Highlights

- Used NNI to quantify and classify AFO spatial patterns in the MRW.
- Used an LMM approach to assess predictive power of AFO spatial pattern.
- AFO-clustered watersheds had elevated nutrients and water temperatures.
- AFO spatial pattern is likely a predictor of nutrients and water temperature.
- Results may support nutrient mitigation by accounting for AFO spatial patterns.

1. INTRODUCTION

The Maumee River Watershed (MRW) is the largest tributary draining into Lake Erie, spanning approximately 6,600 square miles across Ohio, Indiana, and Michigan (Cousino et al., 2015; Larson et al., 2018). Agricultural land use dominates the basin, accounting for 79% of total land area (Miller and Lyon, 2021, Miralha et al., 2021a) and non-point agricultural sources are the primary driver of nutrient loading from the MRW into the Western Lake Erie Basin (Martin et al., 2021). Since the early 2000s, nutrient runoff has been directly associated with the rise in occurrence of Lake Erie eutrophication, leading to harmful algal blooms (HABs; Michalak et al., 2013). HABs pose severe ecological, economic, and public health risks to Lake Erie. Ecological impacts include increased hypoxia and anoxia leading to fish kills, decreased light penetration leading to food-web structure disruption, toxin production inhibiting reproduction of zooplankton and other algae, and trophic cascades that affect larval fish survival and recruitment (Roberts et al., 2009, Watson et al., 2016). Economically, HABs lead to decreased fishing, swimming, boating, and vacationing to Lake Erie, with estimated annual impacts reaching over \$70 million (NOAA, 2025a). These combined environmental and economic impacts have attracted the attention of multiple regulatory agencies that have established the need to reduce nutrient runoff (i.e., 40% reduction of phosphorus runoff) into Lake Erie (Boegehold et al., 2023, GLWQA, 2016, Martin et al., 2021).

Nutrient runoff mitigation in the MRW has historically focused on row-crop agriculture, and the direct land application of inorganic and organic fertilizers (Kast et al., 2021, King et al., 2018). Individual conservation practices, such as no-till farming and drainage modifications, have been used in the region since the 1980s. However, best management practice (BMP) adoption did not become a coordinated, basin-wide priority until the 2015 Western Lake Erie

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3 89 Basin (WLEB) Collaborative Agreement and the 2016 USDA WLEB Initiative (USDA NRCS,
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5 90 2016). Prominent BMPs implemented since then to improve nutrient retention include modified
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7 91 tillage practices, adjusted fertilizer application rates and timing, cover crops, multi-media filters,
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9 92 drainage water management, and buffer strips (King et al., 2018, Scavia et al., 2017, Smith et al.,
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11 93 2015). Paired with edge-of-field monitoring, these practices have supported more targeted, site-
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13 94 specific approaches to nutrient reduction in the MRW (USDA NRCS, 2022, Bukhari, 2015).
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15 95 While fertilizer application in row-crop agriculture is well documented as a major nonpoint
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17 96 source of pollution in the MRW, other nonpoint agricultural sources remain critically
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19 97 understudied, particularly animal agriculture, which is estimated to account for 50% of all
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21 98 nonpoint agricultural pollution in the United States (US; (NOAA, 2025b).

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26 99 Starting in the 1950s, animal agriculture in the US underwent industrialization (Rossi and
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28 100 Garner, 2014; Mallin and Cahoon, 2003). This process consisted of moving animals into large,
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30 101 confined buildings where they are fed until they are ready to be processed (Mallin and Cahoon,
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32 102 2003). This industrialization is primarily driven by economic feasibility and production
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34 103 efficiency (Rossi and Garner, 2014). Animal feeding operations (AFOs) represent most animal
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36 104 farming facilities in the US (US Code of Federal Regulations, 2025). An AFO confines and feeds
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38 105 animals for 45 days or more in a 12-month period and does not yield crop growth on a regular
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40 106 basis. If an AFO also meets a defined threshold number of animals, which varies by type, and
41
42 107 actively discharges to the waters of the US (WOTUS), it can be defined as a confined animal
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44 108 feeding operation (CAFO) and must acquire a permit through the National Pollutant Discharge
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46 109 Elimination Service (NPDES). Because of their intensive manure and wastewater production,
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48 110 AFOs pose threats to water quality, even if they are not directly discharging into WOTUS,
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50 111 primarily through the mismanagement of waste and storm-driven runoff from waste storage (e.g.,
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lagoons) and land-application areas (Burkholder et al., 2007, Martin et al., 2018, Raff and Meyer, 2022). Proximity of AFOs to intermittent streams and drainage ditches further increases the risk of surface water contamination, even where formal siting restrictions are in place (Brands, 2014). AFOs have been associated with water pollution from heavy metals, trace metals, pharmaceuticals, antibiotics, pesticides, sediments, and pathogens (Burkholder et al., 2007, Raff and Meyer, 2022, West et al., 2011). However, some of the most evident water quality impacts of these operations come from nutrient pollution (Glibert, 2020).

Manure contains high levels of nitrogen (N) and phosphorus (P), which can elevate N and P concentrations in surface water and groundwater through runoff and leaching, respectively (McDowell et al., 2001, Rudko et al., 2023). Prior studies illustrate that watersheds with high numbers of AFOs consistently show high surface water concentrations of ammonia, nitrate, and TN (Brown et al., 2020, Weldon and Hornbuckle, 2006). Additionally, P accumulates rapidly in soil, especially when soils are repeatedly exposed to manure application (Qin et al., 2020). This soil accumulation can increase P concentrations in water through runoff and erosion (Mallin and Cahoon, 2020). Nutrient concerns also stem from manure lagoons, in which high volumes of manure may enter water bodies through flooding or infrastructure failure (Mallin et al., 2015). Between 2005 and 2018, the number of animal farms in the WLEB increased by 42%, while the number of animals increased by 226% (EWG, 2022). Additionally, the amount of P introduced to WLEB from manure is estimated to have increased by 67% during the same period. An increase of an estimated 10,610 tons of phosphorus entering the WLEB each year is attributable to animal farm expansion in the region (EWG, 2019). Despite this information, few to no studies have investigated the environmental impact of AFOs in this region, particularly in the MRW.

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3 134 There is evidence to support a relationship between the spatial organization of AFOs and
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5 135 water quality outcomes on a national scale (Miralha et al., 2022). Miralha et al (2022) found that
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7 136 median concentrations of total phosphorus (TP) and total nitrogen (TN) were 0.6mg/L and
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10 137 0.1mg/L higher at outlets of watersheds with a clustered pattern of animal farms, when compared
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12 138 to watersheds with a dispersed pattern (Miralha et al., 2022). Prior work examining the effect of
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14 139 AFO location on environmental conditions found that land surface temperature (LST) increased
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16 140 significantly over two decades within 15 km of animal farms across North Carolina and
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18 141 Michigan, an effect resembling urban heat islands (Miralha et al., 2021b). LST is a well-
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20 142 documented, catchment-scale predictor of stream temperature (Heddum et al., 2020, McNyset et
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22 143 al., 2015, Philippus et al., 2025), and elevated LST from urban heat islands has independently
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24 144 been linked to elevated stream temperature (Somers et al., 2013, Zahn et al., 2021). These
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26 145 findings suggest a plausible pathway by which AFO presence could elevate stream temperature
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28 146 via localized LST increases. However, this pathway has not been directly tested, and no study
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30 147 has examined whether AFO clustering presence affects stream temperature on a regional scale.

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36 148 Hence, the main objective of this study was to assess and predict the relationship between
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38 149 the spatial organization of animal farms and stream water quality outcomes at a local watershed
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40 150 scale (HUC10) within the MRW from 2000 to 2020. We hypothesized that AFO-clustered
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42 151 watersheds would yield higher total phosphorus (TP) and total Kjeldahl nitrogen (TKN)
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44 152 compared to AFO-dispersed watersheds. Additionally, we hypothesized that spatial clustering
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46 153 would compound the localized LST increases already documented around individual AFOs into a
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48 154 stronger, watershed-scale thermal signal, leading to elevated water temperatures (T_w) when
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50 155 compared to a spatially dispersed AFO pattern. To test these hypotheses and quantify the
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52 156 predictive significance of spatial arrangement alongside broader landscape composition, we
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utilized a linear mixed-effects modeling (LMM) approach to isolate the unique impacts of spatial configuration on watershed-scale water quality outcomes. This study aims to fill a gap in the literature by quantitatively testing how AFO spatial clustering shapes nutrient and thermal water quality outcomes at the watershed scale, offering a transferable spatial-statistical framework that can inform targeted AFO siting and watershed management policy beyond the MRW.

2. METHODOLOGY

2.1. Study Area

The study area is the Maumee River Watershed (MRW), located primarily in northwestern Ohio (Fig. 1). The MRW is a transboundary watershed, spanning Ohio, Indiana, and Michigan. The watershed extends from 40° 23' to 42° 5' N latitude and from 83° 20' to 85° 15' W longitude (Verma et al., 2015). The elevation of MRW varies, with the highest elevation areas occurring in the northern portion near Michigan, roughly 1,000 feet above mean sea level (Partners for Clean Streams, 2005). At the Ohio-Indiana border, the elevation is about 707 feet above mean sea level (ODNR, 2001). The lowest elevation is at the mouth of the watershed, roughly 575 feet above sea level (BGSU, 1968). The terrain is relatively flat, with elevation decreasing from the northwest to the eastern portion of the watershed. Soils in the MRW are derived from glacial till material and lacustrine deposits (OH EPA, 2020). This geological history has created soils in the MRW that are deep and range from somewhat poorly to very poorly drained (OH EPA, 2020). Clay surface soils dominate the MRW region, making up about 35-60% of surface soils (OH EPA, 2020). Potential for runoff and soil erosion is high due to low permeability of the clay surface soil. The average annual temperature in MRW is 11.3 °C, with

February (-6 °C) and July (23.6 °C) as the coldest and warmest months. Annual precipitation in MRW ranges from approximately 833mm to 1,135mm. As climate change progresses, this region could experience an increase in temperature from 2-6 °C by the end of the century, and spring and winter precipitation could increase by 20-30% (Cousino et al., 2015). The MRW has approximately 16,000 miles of drainage ditches, developed in the 1850s to drain the Great Black Swamp (OH EPA, 2020). In addition to this considerable surface drainage, there is also considerable subsurface drainage.

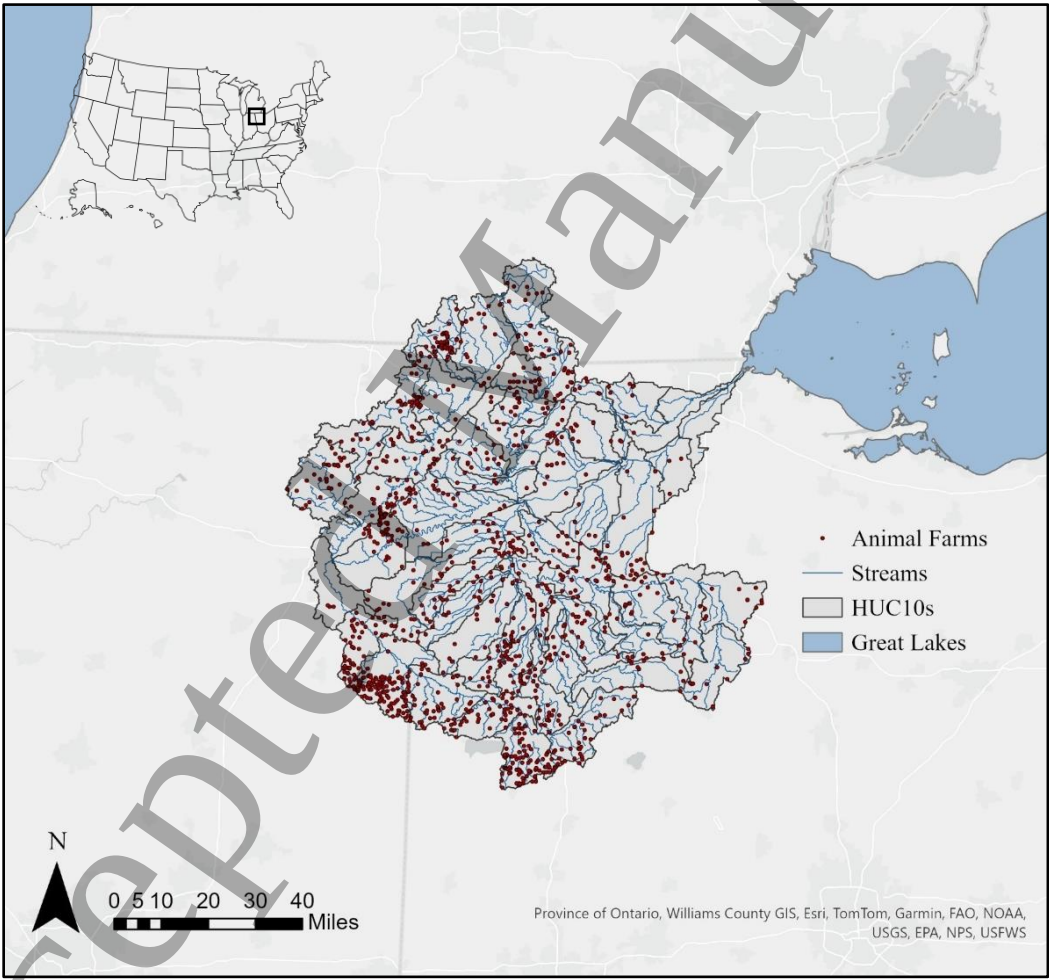


Figure 1 Location of the Maumee River Watershed (MRW), including relevant HUC10 level watershed boundaries, water quality sampling points, major stream networks, and animal farm locations.

2.2. Watershed Boundary and Water Quality Data

The watershed boundary data used for this study were accessed through the United States Geological Survey's (USGS) Watershed Boundary Dataset (WBD). The WBD is a seamless hydrological unit dataset representing hydrological units on a national level. Hydrologic Unit Codes (HUCs) are a two-digit system developed by USGS where each successively smaller watershed is identified by increasing the even-digit number associated with the watershed, ranging from HUC2 to HUC16 (USGS, 2025). HUC10 level watersheds were used in this study (Fig. 1), a preferable HUC size level for agricultural water quality management (McLellan et al., 2018).

All water quality data used for this study were accessed from the National Water Quality Monitoring Council's (NWQMC) Water Quality Portal (WQP), which integrates publicly available water-quality data from the USGS, United States Environmental Protection Agency (US EPA), and over 400 federal, state, tribal, and local agencies (NWQMC, 2025). Physical and chemical water quality data were acquired at the state level for Indiana, Michigan, and Ohio from 2000 to 2020. We selected TP, TKN, and T_w primarily because of data availability. A total of 1,007 stations were located within the MRW region and provided the TP, TKN, and T_w data that were curated for the analysis. After acquisition, we cleaned and sorted all data by location (stream), year, and season using the Python package *pandas*. Season was defined as Mar/20-Jun/20 for spring, Jun/21-09/22 for summer, Sep/23-Dec/20 for fall, and Dec-21-Mar/19 for winter. Water quality data were then spatially referenced using the Python package *geopandas* and joined with HUC10 watershed boundary data (Fig. 1) that displayed a significant AFO spatial pattern. Table S1 shows data availability for the entire MRW by year and the number of samples per analyte.

2.3. Animal Farm Data

Animal farm location data were acquired from the Environmental Working Group (EWG). This dataset includes animal farm type, total animal number per farm, permitted status, and more (EWG, 2022). Animal farm types included poultry, swine, cattle, beef, and dairy. We considered only animal farm locations in watersheds that showed a significant spatial clustered or dispersed pattern in this study. Although we did not include animal farm type in our analysis, HUC10s with significant spatial patterns were dominated by cattle, dairy, and beef operations. Most swine and poultry farms were in locations with no significant spatial pattern (Table S3). For simplicity, the term AFO was used to refer to both AFOs and CAFOs. However, it is important to note that there is a caveat to the federal level regulation about AFO terminology and enforcement per individual states (Alliance, 2025).

2.4. Land Use Analysis

Land use analyses were performed using the National Land Cover Database (NLCD). Annual land cover was downloaded from the Multi-Resolution Land Consortium (MRLC) for the years 2000 and 2020. All “Developed” land use types were combined to simplify analyses. First, the percentage of each land use type was calculated for each year using ArcGIS Pro Version 2.8. Next, we subtracted 2020 land use percentages from 2000 land use percentages to determine land use changes. We performed this analysis (1) to check for major land use transitions, which likely impact water quality responses and (2) to gather land composition data for the modeling step of this study. We applied Mann-Whitney U tests with Benjamini-Hochberg FDR correction to test for significant differences in land use change between dispersed and clustered watersheds.

2.5. Statistical Analysis for Nutrients and Water Temperature

2.5.1. Nearest Neighbor Index (NNI)

NNI calculates the ratio between the observed distance among point features and the expected distance based on the total area within which the point features are located (Clark & Evans, 1954). The observed distance is the average distance between each point and its nearest neighbor (ESRI, 2025). In this study, we used the Average Nearest Neighbor analysis (Eq.1) package in ArcGIS Pro 3.5 to perform the NNI calculation among AFOs within each HUC10. If the observed mean distance is less than the expected mean distance, NNI will be <1 , and the AFOs within the HUC10 area will be classified as “clustered” (i.e., AFO-clustering or clustered watershed). If the observed mean distance is greater than the expected mean distance, NNI will be >1 , and the AFOs within the HUC10 area will be classified as “dispersed” (i.e., AFO-dispersion or dispersed watershed). This pattern is validated by converting the z -score to a p -value, indicating whether the classified pattern is significant. Rigid adherence to $p = 0.05$ has been increasingly challenged in ecological and environmental research (Mudge et al., 2012a, 2012b, Bangdiwala 2016, Palesch 2014, Tomer et al., 2009). Studies have argued that a moderately relaxed threshold would best prioritize statistical power over strict Type I error control, reducing the risk of missing real spatial clustering (Type II error; Mudge et al., 2012a, 2012b). Therefore, we set the significance level to 0.10 ($p < 0.10$) for this work. We also selected only HUC10s with more than two AFOs to account for distance calculation. This analysis evaluated whether significant clustering or dispersion patterns of AFOs were associated with different water quality outcomes within the MRW.

$$ANN = \frac{\bar{D}_O}{\bar{D}_E} \quad \bar{D}_O = \frac{\sum_{i=1}^n d_i}{n} \quad \bar{D}_E = \frac{0.5}{\sqrt{n/A}} \quad \text{Equation (1)}$$

We evaluated median and mean statistics to assess significant differences in water quality (TP, TKN, and T_w signatures) outcomes across spatial patterns (Clustered vs. Dispersed), using a Kruskal-Wallis test and a two-sample t-test with 95% confidence intervals. All significance values were set at 0.05 ($p < 0.05$). A Kruskal-Wallis test is a nonparametric statistical test used to evaluate if groups that fail to meet the normality assumptions of analysis of variance differ on a single variable (Kruskal and Wallis, 1952). A two-sample t-test is a parametric test used to compare the means of two groups (Cressie and Whitford, 1986).

2.5.2. Linear Mixed Effects Model (LMM)

For the models, we only used water quality station data that fell within HUC10s with significant spatial patterns. For T_w , we had 320 stations that provided data almost every season, except for a few winters in some watersheds (Table S4.1). T_w data were well distributed through the stream network, with 15%, 53%, and 32% of the data located in streams of order 1-2, 3-5, and 6-8, respectively. From these streams, 60% were classified as creeks, while 15% were classified as Ditches, 22% as rivers, and 3% as runs. For TP, data availability was reduced significantly (Table S4.2); we had a total of 183 stations distributed throughout the stream network, where 19%, 44%, and 37% of the stations were located in streams of order 1-2, 3-5, and 6-8, respectively. In terms of stream features, most samples were collected in creeks (49%) and rivers (30%), while 19% were collected in ditches and 2% in runs. Although TKN had a similar station distribution to TP (Table S4.3), only 11 of the 20 watersheds with significant spatial patterns had TKN data available, so we ran the models only for TP and T_w .

We fitted a linear mixed-effects model (LMM) algorithm (*statsmodel* python package) to predict TP concentrations and T_w (i.e., dependent variable). The LMM differs from linear models by treating all effects not taken into account by the predictor variables as a random variable. We

chose LMM because (a) its ability to account for intra and inter annual variability and the explicit spatial pattern per watershed in our data (i.e., repeated measures); (b) it can handle fixed and random effects; (c) categorical and continuous variables can be easily accommodated; and (d) the probability of Type I error occurring is low (Jashami et al., 2024, Lessels and Bishop, 2013).

The LMM equation is shown in Eq. (2):

$$y_{ij} = \beta_0 + X_{ij} \beta + t_i + \varepsilon_{ij}, \quad t_i \sim N(0, \sigma_b^2), \quad \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2) \quad \text{Equation (2)}$$

where β_0 is the intercept, X_{ij} is a row vector of predictors (i.e., season, land use [% cropland, wetland, or forest], spatial pattern [clustered and dispersed]) for that specific i^{th} year and j^{th} observation and β is a column vector of corresponding slopes, t_i is the random intercept of the i^{th} random effect variable (year) that is following a normal distribution with zero mean and variance σ_b^2 , and ε_{ij} is the error. Therefore, t_i and ε_{ij} are assumed to be independent. Note that we added a second random intercept to account for the number of farms that was constant per watersheds across years in the dataset. Our modeling goal was to determine the significance of the animal farm spatial pattern (Clustered vs Dispersed) and animal farm amount within a HUC10 in predicting TP and T_w concentrations across MRW. We added catchment physiography variables in the model to investigate if the influence of AFO spatial pattern and presence in predicting TP is significantly comparable to other major water quality impacting factors such as land use, seasonality, and annual variability (wet/dry years). We calculated marginal and conditional r -squared (r^2), root mean squared error (RMSE), and the seasonal correlation (r) between predicted and observed values to investigate the model performance. Final model equations are included in the Supplementary Information as Eq. S1 and S2.

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3. RESULTS

3.1. AFO Clustering and Dispersion

Of the 49 HUC10 watersheds in MRW, 11 displayed significant AFO-clustering patterns (22%), and 9 displayed significant AFO-dispersion patterns (18%) when evaluated at a $p<0.10$ value (Fig.2). Most HUC10s with significant spatial patterns were in the northwestern portion of the watershed, except for one basin on the southeast side. Mapping results also revealed a localized distinction in spatial pattern, where clustered and dispersed HUC10s displayed a neighboring-like behavior. Clustering is a well-observed phenomenon in almost every sector of industrialized production, primarily influenced by economic efficiency (Ayres and Ayres, 2002, Spencer et al., 2010). While clustering may be economically favorable for farmers in the MRW, these localized spatial patterns may reflect proximity to major roads, feed supply networks, or crop fields (Long et al., 2018). Cattle and dairy operations were the primary operations associated with the spatial pattern across the region. Of the 20 watersheds exhibiting statistically significant spatial patterns, 15 were dominated by cattle facilities (6 classified as AFO-dispersed and 9 as AFO-clustered). Poultry operations dominated one AFO-dispersed watershed, while a single AFO-clustered HUC10 watershed showed an even split among cattle, swine, and poultry facilities (Table S3).

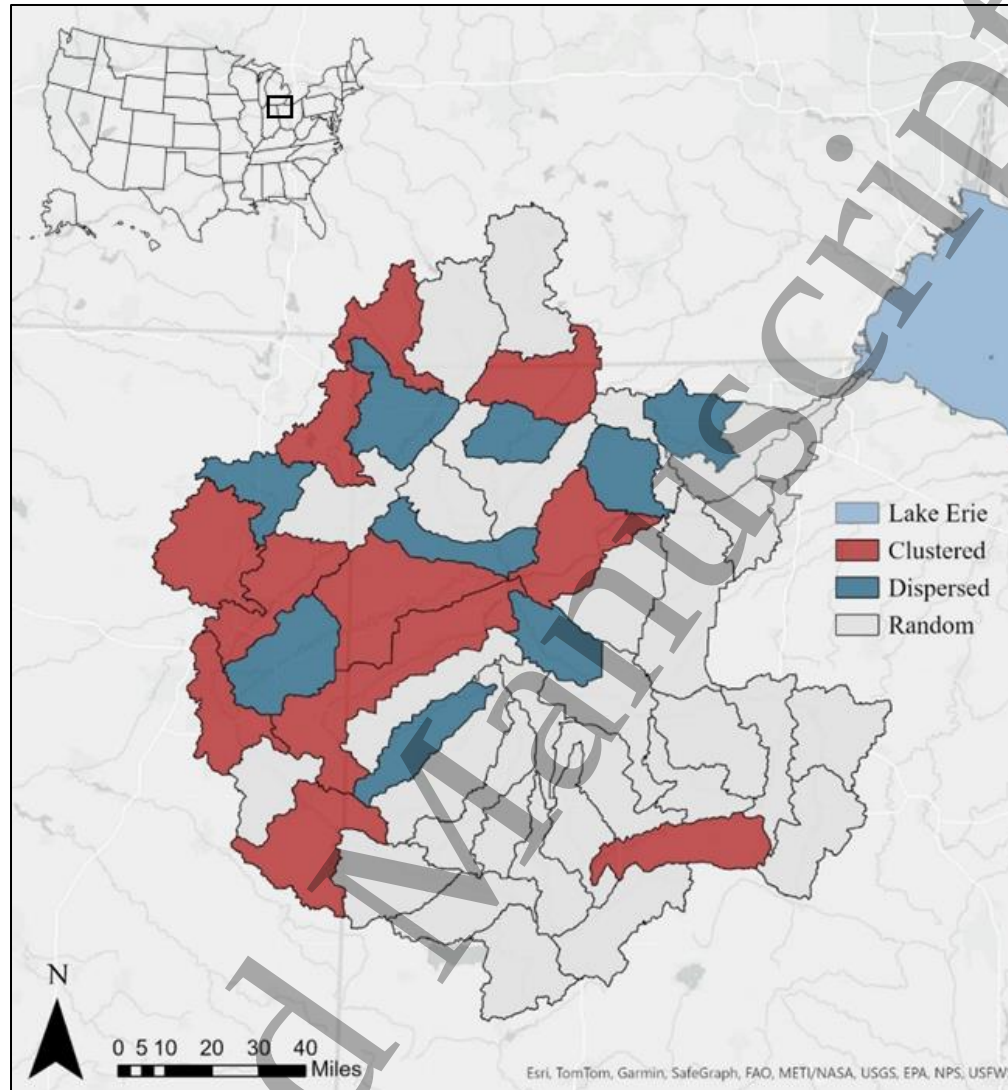


Figure 2. Map of HUC10 spatial organization classification. Areas highlighted in red display a significant clustering pattern of AFOs, while areas highlighted in blue display a significant dispersion pattern of AFOs. Areas highlighted in grey display an insignificant (random) spatial pattern.

3.1.1. Relationship between TP, TKN, and T_w and AFO spatial organization

To evaluate the association between AFO spatial pattern and water quality, TP, TKN, and T_w data were compared across monitoring stations aggregated within AFO-clustered and AFO-

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2
3 336 dispersed HUC10s over the 20-year period (2000-2020). Across all observations, streams
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5 337 draining in AFO-clustered watersheds exhibited significantly higher concentrations of TP and
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7 338 TKN, as well as elevated T_w , compared to those in AFO-dispersed watersheds (p -value < 0.001;
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9 339 Fig. 3). Watershed-level median TP and TKN concentrations were higher in clustered systems by
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11 340 0.05 mg/L and 0.04 mg/L, respectively, while mean concentrations showed larger differences of
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13 341 0.86 mg/L for TP and 0.46 mg/L for TKN (Table 1). Clustered systems were characterized by
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15 342 extreme right-skewed distributions (Fig. S1). TP concentrations in AFO-clustered watersheds
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17 343 reached values exceeding 70 mg/L, whereas peak concentrations in AFO-dispersed watersheds
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19 344 remained below 10 mg/L. While maximum TKN concentrations approached 25 mg/L under both
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21 345 spatial patterns, AFO-clustered watersheds exceeded the 90th percentile value (>3.5 mg/L) at a
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23 346 higher frequency. Across the entire sampling period, watershed-level median and mean T_w were
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25 347 0.60 °C and 1.64 °C higher in AFO-clustered relative to AFO-dispersed watersheds (p <0.001;
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27 348 Fig. 3). Clustered systems maintained elevated thermal baselines across most of the distribution,
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29 349 with a 25th percentile of 10.9 °C (3.6 °C higher than AFO-dispersed systems at 7.3 °C) and a
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31 350 maximum T_w reaching 35.5 °C (4.1 °C higher than AFO-dispersed watersheds; Fig 3).
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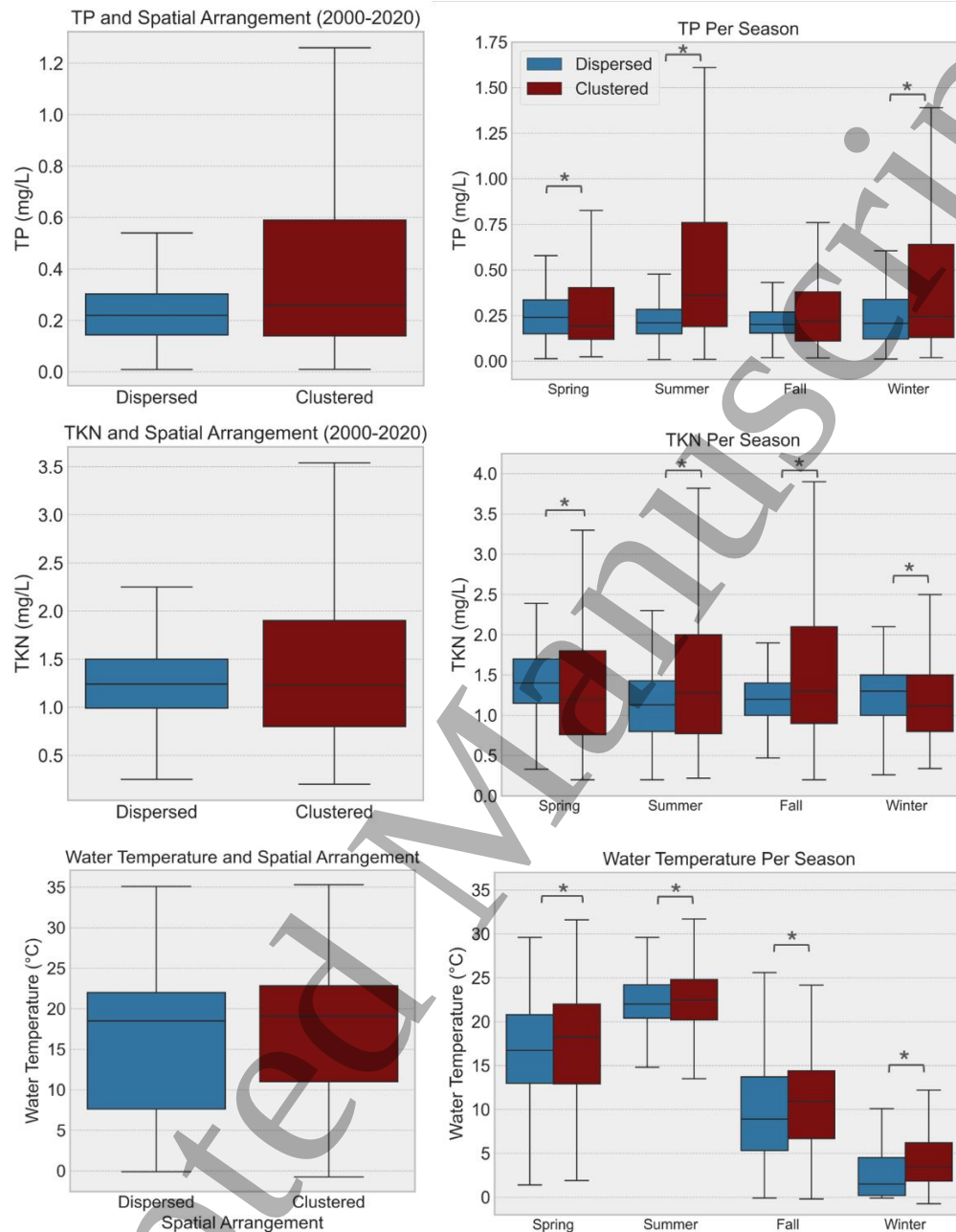


Figure 3. Overall and seasonal total phosphorus (TP), total Kjeldahl nitrogen (TKN) concentrations (mg/L), and water temperature (T_w ; °C) distribution from 2000-2020 per watershed spatial pattern. On the left side of the figure, concentrations and temperature are shown, excluding outliers. On the right side of the figure, the seasonal concentrations and temperature per spatial pattern are shown with a significance level of $p < 0.0001$ indicated as *

Table 1. Full sampling period (Overall) and seasonal mean and median differences in total phosphorus (TP), total Kjeldahl nitrogen (TKN), and water temperature (Tw) between clustered and dispersed watersheds. If positive, clustered watersheds exhibited higher nutrient and Tw than dispersed watersheds. Median values are provided in addition to, and for direct numerical comparison with, the visual medians shown in figure 3.

	Difference (Clustered - Dispersed)					
	TP (mg/L)		TKN (mg/L)		Tw (°C)	
	Mean	Median	Mean	Median	Mean	Median
Overall	0.84	0.04	0.42	-0.01	1.64	0.6
Spring	0.02	-0.05	0.1	-0.2	1.07	1.53
Summer	1.8	0.15	0.63	0.15	0.48	0.46
Fall	0.11	0.02	0.52	0.1	1.21	2.03
Winter	0.2	0.04	0.03	-0.18	1.56	1.9

Seasonal comparisons revealed distinct temporal dynamics between the two spatial patterns (Fig.3). Median TP concentrations differed significantly ($p < 0.001$) across all seasons except fall. The greatest divergence occurred during summer, where median TP in AFO-clustered watersheds exceeded dispersed watersheds by 0.15 mg/L (Table 1). Seasonal mean TP concentrations were consistently higher ($p < 0.001$) in AFO-clustered watersheds throughout the year, peaking in summer at 2.05 mg/L compared to 0.26 mg/L in AFO-dispersed watersheds—a 1.80 mg/L difference. High mean TP concentrations in AFO-clustered watersheds persisted into winter months (0.50 mg/L vs. 0.24 mg/L in AFO-dispersed systems). In spring, median TP output in AFO-dispersed watersheds was slightly higher (0.05 mg/L higher; Table 1) than in AFO-clustered watersheds. However, the significantly high mean values in clustered systems during spring reflect intermittent, high-magnitude phosphorus pulses—a P response also found by other studies in our region (Williams et al., 2016).

For TKN, median concentrations differed significantly ($p < 0.001$) across all seasons, with AFO-dispersed watersheds exhibiting slightly higher median values in spring and winter (Fig. 3). However, mean TKN concentrations remained significantly higher ($p < 0.001$) in AFO-clustered watersheds across every season. Mean TKN concentrations in AFO-clustered watersheds peaked during summer (1.81 mg/L higher than AFO-dispersed) and fall (1.76mg/L higher than AFO-dispersed), demonstrating a seasonal accumulation from spring through fall before declining in winter. Conversely, TKN concentrations in AFO-dispersed watersheds declined from spring through fall before slightly increasing in winter.

Seasonal thermal dynamics (T_w) significantly differed between spatial arrangements across all four seasons ($p < 0.001$; Fig. 3). Seasonal median differences in T_w between AFO-clustered and AFO-dispersed watersheds were 1.53 °C (spring), 0.46 °C (summer), 1.90 °C (fall), and 2.04 °C (winter; Table 1). Mean T_w differences consistently exceeded 1 °C during spring, fall, and winter, indicating that streams in AFO-clustered watersheds experience year-round elevated thermal regimes compared to their AFO-dispersed counterparts. While results indicate a statistically significant difference in thermal signature between AFO-clustered and AFO-dispersed systems during summer, the thermal distribution among spatial arrangements appears to converge during that time of the year. This seasonal pattern—small divergence in summer, larger divergence in spring, fall, and winter—is consistent with the thermal buffering effect of groundwater discharge, which tends to cool streams in summer and warm them in winter relative to surface-runoff dominated flow (Kurylyk et al., 2026). Baseflow contributes substantially and variably to streamflow across the WLEB, with its relative contribution shaped by land use and drainage characteristics (Song et al., 2022), suggesting that differences in baseflow and

groundwater contribution between AFO-clustered and AFO-dispersed watersheds may help explain the observed distributional asymmetry in thermal regimes found in this study.

3.2. Land Use Change per Spatial Pattern

Results indicated an evident trade-off between agricultural and urban land use in the MRW (Fig. 4; error bars represent the 95% confidence interval (CI) for the mean). AFO-dispersed catchments lost $3 \pm 1.5\%$ (mean $\pm$ 95% CI) cropland coverage but gained $2.2 \pm 1\%$ urban and $0.8 \pm 0.4\%$ pasture lands. Similarly, AFO-clustered HUC10s lost $2.7 \pm 0.8\%$ croplands and 0.2% shrublands and gained $1.6 \pm 0.9\%$ urban and $1 \pm 0.5\%$ pasture lands. While these magnitudes varied numerically by spatial pattern, none of the land use change differences between AFO-clustered and AFO-dispersed were statistically significant (Mann-Whitney U tests with Benjamini-Hochberg FDR correction; all-corrected $p > 0.05$). Shrub land use showed the only marginal raw difference ($p=0.028$) but did not remain significant after correction for multiple comparisons ($p > 0.05$). Dispersed watersheds had higher overall agricultural land use ($4.0\% \pm 1\%$) compared to clustered watersheds (Fig. S2); the same was observed for developed land use ($3.2\% \pm 1\%$ higher in AFO-dispersed watersheds). Additionally, forest and wetland land cover were higher in AFO-clustered systems than in AFO-dispersed watersheds (Figure S2). This gap was already present at the start of the study period and remained unchanged through 2020, consistent with the lack of significant land use change detected in either spatial pattern over the study period (Fig. 4). Land use change did not differ significantly between AFO-clustered and AFO-dispersed watersheds. These results suggest that neither land use composition nor its trajectory over time can explain the water quality patterns observed above and highlight the relevance of assessing not only the role of AFOs but also other landscape features that may impact the hydrology and thermal regimes in these watersheds.

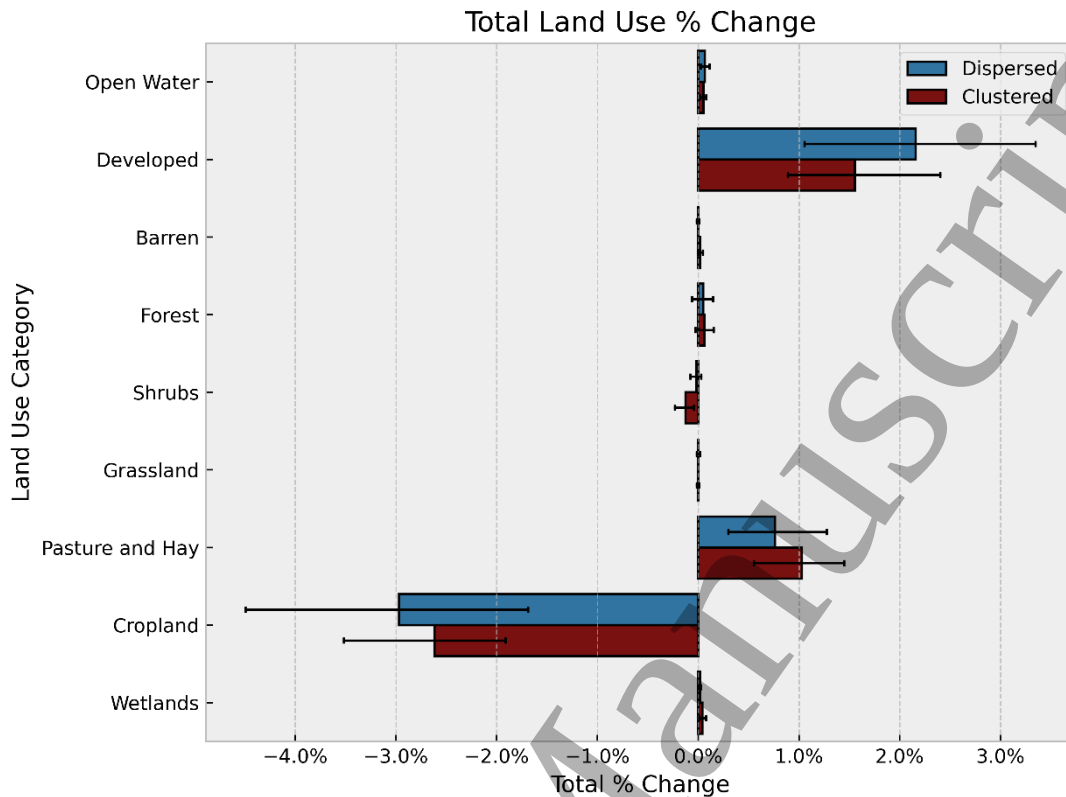


Figure 4. Land use percentage changes from 2000 to 2020 per watershed spatial pattern.

3.3. LMM results

For TP, the dependent variable was log-transformed prior to analysis to comply with model assumptions. Because TP was log-transformed, the model coefficients reflected multiplicative changes in the original scale (mg/L). To account for structural dependencies, random intercepts were included for number of farms per watershed ($\# \text{ farms}$ var = 0.019) and years (Year var = 0.066; Table 2a).

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Table 2. Summary of Estimated LMM Models for (a) Total Phosphorus (TP) and (b) water temperature (T_w)

(a) Response: log10(TP)				
Variable	Levels	Estimate	Std.Error	p-value
Intercept	-	-1.391	0.17	<0.001
Pattern	Clustered	Base		
	Dispersed	-0.117	0.071	0.097
Season	Fall	Base		
	Spring	0.109	0.004	<0.001
	Summer	0.203	0.003	<0.001
	Winter	0.091	0.005	<0.001
Pattern* Season	Clustered*Dispersed*Fall	Base		
	Clustered*Dispersed*Spring	-0.047	0.011	<0.001
	Clustered*Dispersed*Summer	-0.144	0.01	<0.001
	Clustered*Dispersed*Winter	-0.119	0.011	<0.001
Wetland %		-0.043	0.006	<0.001
Cropland %		0.011	0.002	<0.001
Year	random effect	0.066	0.093	-
# of farms	random effect	0.019	0.022	-
Marginal R-squared (Fixed Effects):		0.31		
Conditional R-squared (Full Model):		0.65		
Residual Std Dev (Log scale RMSE):		0.30		

(b) Response: Tw				
Variable	Levels	Estimate	Std.Error	p-value
Intercept	-	10	0.357	<0.001
Pattern	Clustered	Base		
	Dispersed	-0.29	0.033	<0.001
Season	Fall	Base		
	Spring	7.11	0.033	<0.001
	Summer	12.23	0.03	<0.001
	Winter	-6.03	0.044	<0.001
Wetland %		-0.01	0.003	<0.001
Forest %		-0.016	0.004	<0.001
Year	random effect	2.635	0.201	-
Marginal R-squared (Fixed Effects):		0.68		
Conditional R-squared (Full Model):		0.72		
Residual Std Dev (RMSE):		4.3		

Under baseline conditions (Clustered pattern during fall), the predicted geometric mean (intercept) of TP was 0.041 mg/L ($10^{-1.391}$). The model revealed a significant interaction between spatial patterns and seasons ($p < .001$), indicating that the impact of a spatial pattern depends heavily on time of year. Results suggested that transitioning from a clustered to a dispersed pattern yields a suggestive reduction in TP ($\beta = -0.117$, $SE = 0.071$, $p = .097$), approximately 23.6% less TP than in clustered watersheds during fall. However, the significant negative interaction terms (*Pattern*Season*) demonstrated that a dispersed pattern significantly decreases TP concentrations relative to a clustered pattern during all other seasons: *spring* ($\beta = -0.047$, $SE = 0.011$, $p < .001$), *summer* ($\beta = -0.144$, $SE = 0.010$, $p < .001$), and *winter* ($\beta = -0.119$, $SE = 0.011$, $p < .001$). Specifically, relative to clustered watersheds in the same season, AFO-dispersed watersheds yielded roughly 31.5% less TP in *spring* ($\beta = -0.117 + (-0.047) = -0.164$; $10^{-0.164} = 0.685 - 1 = 0.315$), 45.2% less TP in *summer* ($10^{-0.261} = 0.548 - 1 = 0.452$), and 41.9% less TP in *winter* ($10^{-0.236} = 0.581 - 1 = 0.419$). Under clustered conditions, TP concentrations were significantly higher during *spring* ($10^{0.109} = 1.285$, +28.5%; $p < .001$), *summer* ($10^{0.203} = 1.596$, +59.6%; $p < .001$), and *winter* ($10^{0.091} = 1.233$, +23.3%; $p < .001$) relative to fall. Environmental covariates demonstrated significant independent scaling relationships. Every 1% increase in *Wetland* cover was associated with a 9.4% reduction in TP concentrations ($10^{-0.043} = 0.906$, $p < .001$). Conversely, each 1% increase in *Cropland* cover was associated with a 2.6% increase in TP ($10^{0.011} = 1.026$, $p < .001$). Overall, fixed effects explained 31% of the variance in TP (marginal $r^2 = 0.31$), while adding *Year* and *# of farms* per HUC10 as random effects brought the total explained variance to 65%. The residual standard deviation (log-scale RMSE) of the model was 0.30, indicating a multiplicative error factor of approximately 2 ($10^{0.30}$). Figure 5 shows the fit between observed and predicted TP per season.

The seasonal correlation between observed and predicted values was consistently high with our fitted model, where r was 0.68 (fall), 0.63 (spring), 0.71 (summer), and 0.53 (winter).

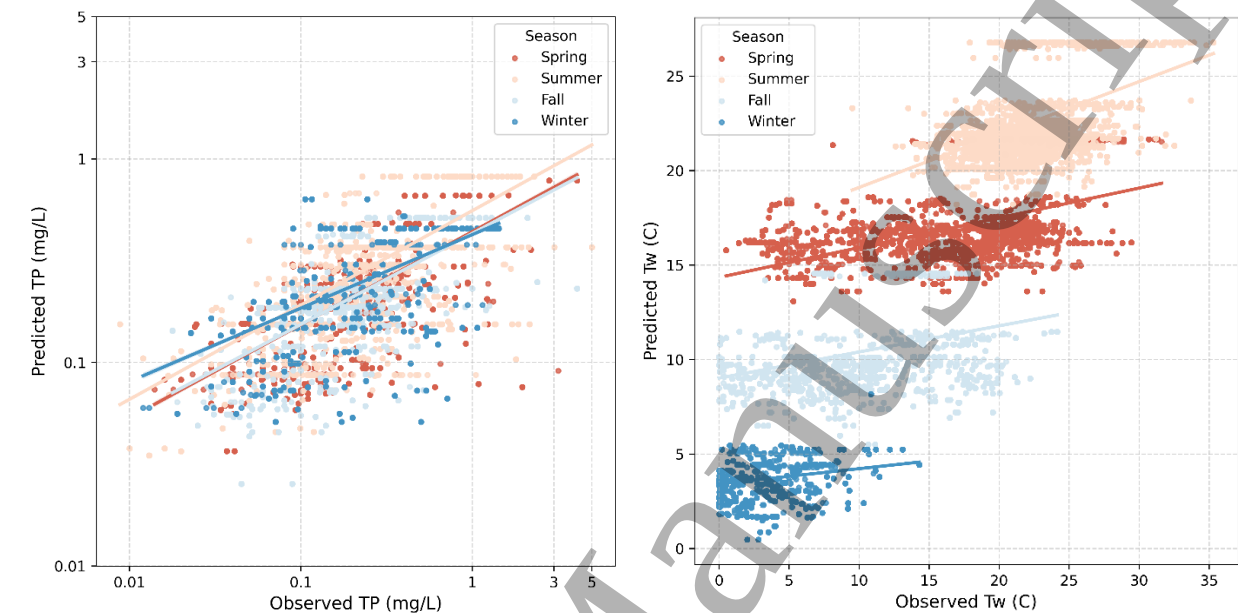


Figure 5. Graphs illustrate the observed versus predicted model fit for Total Phosphorus (TP) and Water Temperature (T_w) across seasons.

We fitted a similar model for T_w . However, we removed the interaction term because T_w seasonal progression (Fig. 5) did not interact with the spatial pattern. While we added the # of farms as a random effect for TP, it did not contribute to the variance in T_w and was omitted. Overall, the fixed environmental predictors (i.e., *Pattern*, *Season*, *Wetlands*, *Forest*) accounted for 68% of the total variance in water temperature (Marginal $r^2 = 0.68$). Adding the random effect of *Year* increased the total explained variance to 72% (Conditional $r^2 = 0.72$) with a residual standard error of 4.3°C.

All fixed effects were highly statistically significant ($p < 0.001$; Table 2b). Relative to the baseline reference configuration (clustered spatial pattern during the *fall*; Intercept = $10 \pm 0.36^\circ$ C), watersheds characterized by a dispersed pattern exhibited significantly lower T_w across the

study period ($\beta = -0.29 \pm 0.03^\circ\text{C}$). As expected, season exerted a strong influence on thermal dynamics: stream temperatures increased by $7.1 \pm 0.03^\circ\text{C}$ in *spring*, peaked at an additional $12.2 \pm 0.03^\circ\text{C}$ during *summer*, and dropped by $6.0 \pm 0.04^\circ\text{C}$ in *winter* relative to *fall*. Both land-use variables provided statistically significant thermal buffering and cooling effects; every 1% increase in wetland and forest coverage lowered stream temperatures by $0.1 \pm 0.003^\circ\text{C}$ and $0.016 \pm 0.004^\circ\text{C}$, respectively. Finally, considerable inter-annual variation was captured by the random effect *Year* ($\sigma_{\text{Year}}^2 = 2.635 \pm 0.2^\circ\text{C}$), validating the need to control for temporal climatic shifts across the 20-year dataset.

4. DISCUSSION

4.1 TP and Manure Dynamics in MRW

Our findings indicate that AFO clustered watersheds exhibit significantly higher nutrient loading than AFO dispersed watersheds. TP concentrations in clustered watersheds exceeded the TP targets (0.23mg/L) determined by U.S. EPA's Action Plan for Lake Erie in every season examined (US EPA, 2018). Notably, summer TP concentrations in clustered watersheds were 9 times higher than this threshold. This is likely due to land application of both inorganic fertilizers and manure, which leads to N and P exceedances (Liu et al., 2017, Long et al., 2018). In the MRW, liquid manure is most often applied from April to October, with nearly 80% of all planned manure applications falling within this window (Kast et al., 2019). Our TP model supports the role of AFO spatial organization in driving TP patterns across the MRW. The evidence for an AFO-clustering effect on TP comes from the significant *Pattern* $\times$ *Season* interaction terms, which show that the dispersed-vs-clustered TP difference strengthens considerably in spring, summer, and winter (31.5%, 45.2%, and 41.9% lower TP in dispersed watersheds, respectively) while remaining statistically indistinguishable from the fall baseline. This seasonal pattern is

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3 495 consistent with the manure application dynamics described in Kast et al (2019). Approximately
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5 496 78% of all manure P generated by permitted operations in the MRW was reported as allocated to
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10 498 However, liquid and solid manure are almost always land applied within the same watershed
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12 499 containing the AFO that produced it. Even when manure is hauled off-site, farmers typically
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14 500 transport liquid manure only 2-9 miles before land application in the MRW(Kast et al., 2021,
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16 501 Long et al., 2018). Additionally, 65% of wet manure volume was planned to be applied to fields
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18 502 under the direct control of AFOs that produced it (Kast et al., 2019). Together, these patterns
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20 503 suggest that wet manure produced by clustered AFOs is likely land applied within the same
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22 504 watershed in which it was generated, potentially compounding nutrient loading in the already
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24 505 agriculturally dominant, AFO-clustered systems.
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29 506 TP concentrations were unexpectedly high in winter months for both clustered and
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31 507 dispersed watersheds. According to the Ohio Revised Code 905.326 and 939.08, manure in the
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33 508 MRW may not be applied 1) on snow-covered or frozen soil, 2) when the top 2 inches of soil are
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35 509 saturated from precipitation, or 3) when the local forecast states greater than 50% chance of
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37 510 precipitation exceeding 1-inch within 24 hours (Ohio Revised Code, 2016). Expected results
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39 511 would yield the lowest overall TP output in winter months, given low temperatures, minimal
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41 512 inorganic fertilizer and manure application, and reduced storm-driven runoff (US EPA, 2023a).
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43 513 This unexpected pattern suggests several potential causes, particularly manure application on
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45 514 frozen, snow-covered soil, which can generate high phosphorus runoff (NRCS, 2020, Owens et
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47 515 al., 2011). This winter TP signal may also reflect legacy phosphorus stored in agricultural soils
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49 516 and tile drainage systems from decades of prior fertilizer and manure application in the region.
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54 517 Edge-of-field monitoring in Ohio has shown that legacy ('old') soil phosphorus, rather than
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recently applied ('new') fertilizer or manure, dominates dissolved reactive phosphorus losses through subsurface tile drainage, accounting for the majority of phosphorus loss across monitored sites (Osterholz et al., 2023). This legacy signal may be compounded by seasonal timing. In Canadian Lake Erie tributaries, tile-drainage proportion was positively associated with phosphorus loading, especially during the thawing season, when frozen soils begin to release stored phosphorus as they thaw (Tedeschi et al., 2024). These seasonal trends and links to manure application warrant further investigation, as previous literature describes avoiding winter manure applications at permitted animal farms in the MRW, suggesting significant nutrient output from undocumented operations (Kast et al., 2019, Meyer et al., 2024).

4.2. TKN Dynamics

The divergence between TKN's median and mean patterns is informative rather than contradictory. Median concentrations reflect the most frequently observed TKN values in a watershed and are relatively insensitive to a small number of extreme observations. Conversely, mean values are pulled upward by even infrequent high-magnitude events. The fact that AFO-dispersed watersheds showed slightly higher median TKN in spring and winter, while AFO-clustered watersheds showed significantly higher mean TKN in every season, suggests that typical, baseline TKN concentrations are not consistently elevated in AFO-clustered watersheds, but episodic, high-magnitude TKN pulses are. This mirrors the pattern observed for TP, suggesting that not only phosphorus pulses may be tied to manure application timing (Williams et al., 2016). This mechanism may also be plausible for TKN, given that organic forms of nitrogen and ammonia are closely associated with manure application (Diaz et al., 2010). This is particularly relevant in the MRW. Although total nitrogen loads to the Maumee River have remained largely consistent over recent decades, the proportion of TKN has been increasing

relative to oxidized inorganic nitrogen (Newell et al., 2019). Long-term TKN trend analysis in the MRW (1982-2022) indicated that TKN significantly contributes to algal bloom extent predictions when considered jointly with soluble reactive phosphorus, total suspended solids, and flow, despite historically receiving less research attention than P in the MRW (Khan et al., 2026). Measuring TKN alongside other N and P parameters may improve our ability to distinguish inorganic and organic N pools and advance our understanding of the role of AFOs in N pollution in the MRW. Using TKN may also inform watershed modeling of N pools and processes in AFO-dominated watersheds.

4.3 Water Temperature (T_w)

Water temperature was higher both seasonally and over the entire study period in AFO-clustered watersheds. Notably, fall and winter median T_w in AFO-clustered watersheds were approximately 2° C higher than in AFO-dispersed watersheds. T_w is one of the most influential factors in stream ecology, and observed increases in T_w can drive changes across biological, chemical, and physical processes (Johnson et al., 2024, Leach et al., 2023). In the MRW, a primary concern is the cascading effect of increased T_w on aquatic ecosystems. Elevated water temperatures can reduce dissolved oxygen and redox potential at the sediment-water interface and stimulate microbial activity, both of which can increase the release of sediment-bound N and P into the water column, potentially intensifying nutrient loading and algal bloom risks (Cheng et al., 2020, Hasan et al., 2023, Barica, 1993, US EPA 2021). This mechanism is best documented in depositional, low-flow, or poorly oxygenated settings, including Western Lake Erie itself, where internal sediment P loading is expected to become more significant as lake sediments warm under climate change (Gibbons and Bridgeman, 2020). Within the MRW's stream network, this process may be most relevant in slower-moving, channelized reaches such as the

ditch-dominated segments most present in our AFO-clustered watersheds (Tables S4), where reduced flow and canopy cover could favor both sediment deposition and elevated water temperature simultaneously. Several models predict that the MRW will undergo increased nutrient loading as climate change progresses, primarily due to increased temperatures and precipitation (Miralha et al., 2021a, Culbertson et al., 2016, Bosch et al., 2014, Seybold et al., 2022). Given the combination of higher nutrient concentrations and elevated T_w observed in this study, the potential ecological consequences under a changing climate are considerable. AFO-clustered watersheds in the MRW appeared to experience lessened temperature buffering, as evidenced by lower minimum, higher maximum, and greater overall seasonal range in stream temperature relative to AFO-dispersed watersheds. This reduced buffering capacity is likely to further exacerbate the factors described above.

Our LMM model confirms that AFO spatial clustering significantly impacts MRW stream thermal regimes (AFO-dispersed T_w 0.29 °C cooler than AFO-clustered systems), even after controlling for seasons and land cover (wetlands and forest). Unlike the TP model, the thermal effect of spatial clustering required no season interaction, indicating that the clustering effect on T_w is structurally constant year-round. While observational differences in T_w fluctuated seasonally (from 0.46 °C in summer to 2.04 °C in winter), the model attributed these apparent shifts to regional temperature cycles (*Seasons*) and land cover, isolating a consistent pattern-specific cooling effect once these controls are applied. This residual seasonal structure is consistent with baseflow buffering dynamics previously reported in the region (Song et al., 2022), though our model did not test baseflow as a covariate. Mechanistically, this season-independent effect aligns with the unshaded, channelized drainage ditches characteristic of clustered AFO-watersheds, rather than transient, seasonally driven processes like manure

application. Given the complexity of these underlying hydrological drivers, further research is warranted to isolate the pathways through which AFO clustering alters local stream microclimate.

4.4 Land Use Dynamics

Quantifying land use cover and historical land use change provided insights into how landscape characteristics influence regional water quality outcomes across the MRW (Kronvang et al., 2020). Agricultural land use decreased and developed land use increased, but these changes did not differ significantly between AFO-clustered and AFO-dispersed watersheds. Furthermore, agricultural land use remained consistently higher in dispersed than clustered watersheds throughout the study period. Despite lower overall agricultural crop density, AFO-clustered streams were warmer and had higher nutrient concentrations. This pattern persisted even after controlling for land cover in the LMM, indicating that AFO clustering predicts water quality outcomes independently of land-use context.

Our total phosphorus LMM model revealed that every 1% increase in wetland cover was associated with a 9.4% reduction in TP concentrations, whereas every 1% increase in cropland was associated with a 2.6% increase. Wetlands provide a myriad of ecological and water quality benefits, and wetland restoration and implementation in agricultural areas is a well-accepted method of nutrient retention (Page et al., 2023, H2Ohio, 2025, Mitsch et al., 2005). However, the ecologically favorable effects of wetlands and forests may be diminished when a watershed has a clustered pattern of animal farms, depending on the location of wetlands and forests within a catchment (Miralha et al., 2022, Tong and Chen, 2002). In other Great Lakes agricultural watersheds, riparian forest cover has been shown to reduce P storage in stream sediments, suggesting that the placement of forest cover relative to the stream network, not just in the

overall watershed extent, likely shapes its effectiveness in the MRW (Kreiling et al., 2021). In addition to nutrient retention, our LMM model revealed decreases in T_w with a 1% increase in forest or wetland cover in AFO-clustered catchments, indicative of the temperature buffer effect these systems can provide (Bowler et al., 2012, Dick et al., 2017). To leverage these well-documented water quality benefits, other US states, such as North Carolina, have already implemented targeted riparian forest buffer policies surrounding AFOs to safeguard water resources near liquid-manure animal facilities (Miralha et al., 2021b).

Collectively, these results demonstrate that focusing solely on row-crop agriculture as the primary driver of nutrient pollution fails to capture the full scope of water quality degradation in the MRW. While coarse-resolution (30 m) land cover data effectively captured these broad relationships, future research utilizing higher-resolution spatial data could further clarify fine-scale landscape interactions in AFO-clustered watersheds. Ultimately, targeted investments in riparian buffer strips and strategic wetland restoration represent essential pathways to support stream and habitat recovery across agriculturally clustered systems.

4.5 Policy Considerations and Limitations

The Ohio EPA has recognized AFOs as a source of nutrient pollution in the WLEB, primarily emphasizing the impact of manure application, citing that manure is responsible for roughly 8% of all P contributions to WLEB (US EPA, 2023b). This 8% metric, however, likely underestimates the actual impact of AFOs on nutrient loading in the region. Our results indicate a direct association between increased TP/TKN concentrations and AFO clustering, suggesting that AFOs may intensify nutrient exports in watersheds dominated by row-crop agriculture such as the WLEB. To our knowledge, no work has investigated the relationship between AFO-clustering and stream warming such as ours. This study, alongside previous literature, calls on government

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agencies and watershed managers to recognize that the impact of AFOs on water quality extends beyond manure application – a distinction that deserves consideration for policy and watershed resilience.

Discrepancies in compliance and oversight may also exist at the state level (US EPA, 2025b). MRW is a transboundary watershed, and the animal farms within this boundary are managed under considerable differences. For example, Indiana regulates animal farms under the Indiana Department of Environmental Management (IDEM). IDEM requires any area designated as an AFO, or confined feeding operation (CFO), to obtain a state permit, regardless of facility size or direct facility discharge (IDEM, 2025). In Ohio, the Ohio Environmental Protection Agency (OhEPA) and the Ohio Department of Agriculture (ODA) split AFO and CAFO regulations. Ohio implements federal-level regulations for CAFOs but also requires specific implementation of manure management plans and certified managers for operations defined as “large” under federal-level regulation (OH EPA, 2025). Michigan’s Department of Environment, Great Lakes, and Energy (EGLE) regulates AFOs and CAFOs statewide (EGLE, 2025). EGLE has the authority to define any AFO as a CAFO under a certain set of criteria that are not explicitly described at the federal level, including general proximity to water bodies and potential for high pollution production (EGLE, 2025). Under Michigan law, CAFOs are point sources and must obtain NPDES permits unless facilities are determined to have no discharge potential (EGLE, 2025). The transboundary nature of the MRW creates complex water quality management circumstances. Although this study does not focus on these specific political discrepancies between states, this factor should be considered in future water quality management in the MRW. Cross-referencing these organizational patterns with manure

application frequencies and volumes could reveal critical relationships and ultimately guide strategies to prevent nutrient overloading in the MRW.

This study verifies a predictive relationship between AFO spatial organization and localized water quality outcomes. These spatial patterns can be integrated into future environmental modeling. Despite the absence of climatic, soil, flow, and elevation variables, our models still explained 31% of TP and 68% of T_w . While variation in TP and T_w output using the LMM approach was high, this variability was expected when modeling nutrient exports in the MRW. TP fluctuations are dependent upon factors, including heavy rainfall, seasonal runoff, streamflow variations, climate, and geomorphology (Alvarez et al., 2009, McDowell et al., 2001, Mitsch et al., 2005, Sharpley et al., 1993). Temperature and precipitation are often described as influential drivers of phosphorus export (Correll et al., 1999, Sabo et al., 2023). Characterized by flat topography and consistently high clay content, the MRW's physical landscape is remarkably uniform. More specifically, clay-rich soils are more likely to experience phosphorus loss during large precipitation events, emphasizing the importance of considering precipitation and flow in these modeling exercises (Djodjic et al., 2000). Although we had access to flow data, few to no stations matched the water quality sample locations. Increasing the spatial and temporal resolution of water quantity and quality monitoring in AFO-clustered watersheds could greatly improve the insights from this work.

5. CONCLUSION

Our multi-decade analysis found that HUC10 watersheds with clustered AFOs had significantly higher TP and TKN, validating national-level analyses. AFO-clustered streams were also consistently warmer, with greater seasonal thermal fluctuation, than AFO-dispersed streams—to our knowledge, the first evidence linking AFO spatial clustering to stream warming

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3 678 in the MRW. Seasonally, TP and TKN peaked in summer, consistent with manure application
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5 679 timing, while winter months yielded high TP and T_w in AFO-clustered watersheds, a pattern that
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7 680 points toward legacy phosphorus and undocumented manure application as priorities for future
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9 681 investigation. The reduced capacity of AFO-clustered watersheds to buffer temperature extremes
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11 682 is particularly concerning given climate trends and the need for resilient coastal systems across
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13 683 the Great Lakes. Our LMM confirmed that AFO spatial pattern strongly predicts TP and T_w
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15 684 outcomes in the MRW, even after accounting for land cover, season, and other landscape
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17 685 variables. Land cover itself remained ecologically important — wetlands in particular show
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19 686 potential to meaningfully reduce TP exports, and forest and wetland cover measurably buffered
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21 687 stream temperature at the local scale — but its benefits appear to depend on placement relative to
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23 688 the stream network at least as much as on overall watershed extent. This distinction, if confirmed
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25 689 by future work, would meaningfully reframe how riparian restoration is targeted in AFO-dense
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27 690 watersheds: prioritizing forest and wetland placement directly along ditch networks, not simply
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29 691 increasing watershed-wide cover. These findings can guide future investigations into how AFOs
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31 692 affect water quality in the MRW. While existing regional research focuses heavily on row-crop
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33 693 agriculture and isolated manure applications, it has yet to fully account for the broader
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35 694 landscape-scale effects of animal farming operations.
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6. ACKNOWLEDGMENTS

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